

KIANA, AK, USA

WMO: 701196

Lat: 66.976N

Lon: 160.437W

Elev: 168

StdP: 14.61

Time Zone: -9.00 (NAK)

Period: 09-19

WBAN: 00102

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.2	-35.8	-34.5	0.8	-28.3	-32.4	0.9	-26.8	23.0	19.9	20.2	17.9	2.3	340	0.759

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	75.4	58.8	72.3	57.9	69.1	56.6	61.0	71.5	59.4	68.6	58.0	66.7	7.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.9	69.4	62.9	55.1	65.1	61.8	54.1	62.5	60.4	27.0	71.2	26.0	68.6	25.0	66.8	65.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.2	16.9	15.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	-3.5	4.0	6.6	24.1	40.4	55.1	57.5	52.2	42.1	28.5	8.5	1.6		
	DBStd	24.88	20.89	16.95	12.85	11.37	9.81	7.03	5.76	6.02	7.67	8.75	14.05	15.06		
	HDD50	9126	1657	1288	1345	776	316	30	8	45	252	666	1244	1499		
	HDD65	14044	2122	1708	1810	1226	762	304	240	396	687	1131	1694	1964		
	CDD50	574	0	0	0	0	19	183	241	115	16	0	0	0		
	CDD65	16	0	0	0	0	0	7	9	0	0	0	0	0		
	CDH74	196	0	0	0	0	3	99	92	2	0	0	0	0		
	CDH80	28	0	0	0	0	0	20	8	0	0	0	0	0		
	WSAvg	6.0	5.7	6.7	5.7	6.1	6.3	6.5	5.6	5.6	5.8	6.2	5.9	5.8		
	PrecAvg	18.3	1.3	1.4	1.0	0.9	0.8	1.1	2.2	2.8	2.0	2.0	1.5	1.4		
(15) Precipitation	PrecMax	23.5	2.7	4.2	3.7	2.5	2.0	1.9	4.6	6.8	3.6	4.3	3.9	3.0		
	PrecMin	13.7	0.2	0.3	0.3	0.2	0.2	0.5	0.8	0.9	0.9	0.6	0.3	0.7		
	PrecStd	2.6	0.7	0.9	0.7	0.6	0.4	0.4	0.8	1.2	0.7	0.9	0.9	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.4	35.8	37.5	52.1	71.8	83.0	81.1	73.4	64.4	50.7	37.8	35.7		
		MCWB	32.3	35.0	32.5	42.1	54.2	61.6	61.2	58.3	52.2	38.7	34.7	32.9		
	2%	DB	31.6	32.6	33.7	47.4	64.3	76.3	75.9	69.2	60.4	45.2	34.3	32.4		
		MCWB	29.7	31.1	30.3	38.9	50.9	57.8	60.0	57.7	50.1	39.4	32.9	30.0		
	5%	DB	28.0	29.7	29.5	43.4	60.1	72.4	72.8	65.7	56.8	42.6	30.6	28.2		
		MCWB	26.6	28.2	27.0	36.4	48.6	56.6	59.4	56.5	49.6	38.3	29.5	26.9		
	10%	DB	23.7	26.8	25.4	39.5	55.4	69.2	69.7	63.0	54.0	39.7	27.0	22.9		
		MCWB	22.2	25.1	22.7	33.7	46.3	55.3	58.3	55.1	48.4	36.6	25.7	21.7		
	0.4%	WB	33.4	34.7	34.1	42.4	54.6	62.1	63.7	61.2	54.4	42.9	35.0	33.2		
		MCDB	34.8	35.2	36.1	50.9	69.5	78.2	77.0	67.8	60.3	44.4	37.0	35.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	30.1	31.7	30.6	39.5	51.6	59.7	61.8	59.3	52.7	40.8	33.0	30.7		
		MCDB	31.5	32.8	33.0	46.8	62.5	71.6	73.0	65.8	57.0	44.0	34.3	32.1		
	5%	WB	27.5	28.3	27.0	36.9	49.5	58.1	60.3	57.9	50.9	39.1	29.8	26.9		
		MCDB	28.9	29.5	29.3	42.6	58.3	69.1	70.2	64.4	54.5	41.9	30.7	28.9		
	10%	WB	23.9	25.4	23.3	34.6	47.4	56.6	58.9	56.2	49.3	37.1	25.8	21.9		
		MCDB	25.6	27.1	25.5	39.1	54.9	66.9	68.0	61.4	53.7	39.5	27.0	23.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	14.1	18.6	17.7	17.1	17.9	14.9	14.1	14.1	10.1	10.9	11.5		
		MCDBR	12.7	11.6	15.9	17.9	23.6	23.6	22.3	18.9	18.0	10.3	10.9	10.6		
		MCWBR	11.5	11.0	12.6	11.2	12.5	10.0	9.5	9.7	10.2	6.6	10.2	9.3		
		MCDBR	13.5	12.0	14.0	16.7	21.0	19.6	19.7	15.5	12.6	8.6	10.8	10.6		
	5% WB	MCWBR	12.3	11.7	11.5	10.8	11.4	8.9	9.0	8.4	7.8	6.4	10.2	9.4		
(40) Clear-Sky Solar Irradiance	taub		0.177	0.238	0.256	0.322	0.355	0.349	0.363	0.358	0.316	0.256	0.171	N/A		
	taud		1.904	2.128	2.251	2.118	2.242	2.419	2.399	2.428	2.490	2.318	1.926	N/A		
	Ebn at Noon		62	193	258	261	265	272	263	251	237	188	60	0		
	Edh at Noon		4	16	24	36	36	31	31	27	20	13	3	0		
(44) All-Sky Solar Radiation	RadAvg		22	185	735	1417	1718	1867	1491	1097	677	245	46	0		
	RadStd		3	24	61	87	181	175	131	146	81	30	7	0		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+2.57	N/A	N/A	+1.80	+1.48	N/A	-855	-918	+117	+2

Nomenclature: See separate page